

Johny Yong Jun Siang

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SUMMARY

Software Developer and B.Sc. (Hons) Interactive Media & Game Development student with over 5 years of experience in **C++, C#, Python (intermediate)**, and **JavaScript**. Strong background in **game systems engineering, multiplayer networking, and custom engine development**, with hands-on experience in **web development using React, HTML, and CSS**. Proven ability to take ownership of both **team-based and solo projects** across the full development lifecycle, from system design to deployment.

NOTABLE PROJECTS

Battlemania

- **Challenged myself to master 2D fighting game mechanics**, independently researching and implementing frame-data logic and precise melee hitboxes.
- **Having learnt Forge Networking, I expanded my networking expertise** by experimenting with the **Photon Networking** package to solve latency issues in high-speed combat scenarios.
- **Adopted a "systems-first" mindset**, focusing on building a robust underlying engine for combat and movement before integrating placeholder art assets.
- **Managed the full development lifecycle** as a solo creator, from initial concept to a functional multiplayer prototype.

Gravity

- Led a 9-member team to develop a physics-driven top-down survival game built on a **custom C++ engine**, managing the product roadmap and technical milestones.
- Designed the engine's **Entity Component System (ECS)**, including a Game Object Factory and a Tag/Layer Manager to optimize object handling and performance.
- Authored comprehensive **Technical Design Documentation** using **MkDocs**, maintaining a live version via GitHub Pages for team alignment and version control.

Developer Portfolio

- Previously created my portfolio on [Wix.com](https://wix.com), I decided to rebuild my personal developer portfolio using **React**, with a focus on clean structure and reusable components.
- Hosted on Github Pages
- Used **HTML, CSS, and JavaScript** to design and style the site, making sure it works well across different screen sizes.
- Set up the project with **Node.js tooling** for local development and builds.
- Organised project pages to clearly explain technical work, design decisions, and implementation details.

EDUCATION

B.Sc.(Hons), Interactive Media & Game Development

Singapore Institute of Technology / DigiPen Institute of Technology

~August 2023 - April 2027

Diploma in Game Development and Technology

Nanyang Polytechnic

~April 2018 - April 2021

SKILLS

Programming & Scripting: C, C++, C#, Python (Intermediate), Java (Intermediate), HTML, Lua, JavaScript.

Networking & Backend: Photon Networking, Forge Networking, Firebase, Azure DevOps, JSON

Software & Tools: Visual Studio 2022, Visual Studio Code, Git, GitHub, Android Studio, Flutter, MkDocs, React, Node.js

Project Management & Design: Technical Design Documentation, Cross-functional Team Leadership, UI/UX Design, Agile Workflow

Spoken Languages: English (Fluent), Mandarin (Fluent)

ACHIEVEMENTS

Edusave Good Progress Award

Nanyang Polytechnic

Commanding Officer's Coin

Singapore Armed Forces

EXPERIENCE

Mentor, Singapore Institute of Technology Community Challenge

Singapore Institute of Technology - Part Time

~October 2025 - October 2025

- I volunteered to mentor teams in using **VittaScience AI** posture & image recognition tools for prototype development.
- Supported the students across 5 days on their programming workflows, feature planning, UI/UX design decisions, and technical troubleshooting.

Auto Technician

Singapore Armed Forces - National Service

~May 2021 - August 2023

- Maintained and repaired military vehicle systems as an auto technician
- Collaborated within a technical team under strict deadlines and safety protocols
- Developed a storage management mobile app during National Service using Azure DevOps & Flutter for development and Firebase for real-time tracking of resource usage.

Software Engineer

YJP Surveyors Pte Ltd - Internship

~June 2020 - August 2020

- Contributed to a HoloLens mixed reality project built in Unity, enhancing visualisation of underground pipes and connections normally hidden by walls or flooring.
- Implemented **voice recognition** features to improve user interaction and accessibility.
- Gained hands-on experience with point cloud data and **LiDar technology** to support spatial mapping and alignment.
- Delivered a fully functional prototype that was commended for its effectiveness and innovation.

CERTIFICATIONS

Playstation Game Development Training Completion

- Successfully completed industry recognised training in Playstation workflows by utilising the Playstation DevKit.

AI For Industry: Literacy in AI

- Gained foundational knowledge of artificial intelligence concepts and their industry applications.